UNIVERSITY INTERSCHOLASTIC LEAGUE

Mathematics

District • 2026

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1. **Jordan earns \$18.25 per hour for the first forty hours each week at his job at Rooftop Solutions. He earns \$27.50 for any hours more than forty, but not exceeding fifty. He earns \$36.50 for any hours more than fifty. If Jordan worked 72 hours last week, how much did he earn?**

   (A) \$1800.00 (B) \$1802.00 (C) \$1804.00 (D) \$1806.00 (E) \$1808.00

2. **Consider line segment $\overline{AB}$ with endpoints $A(-6, 8)$ and $B(9, -7)$. Point $C(2, b)$ lies on the perpendicular bisector of $\overline{AB}$. b = _______.**

   (A) 0 (B) 1 (C) 2 (D) 3 (E) 4

3. **Last summer, Arlene left home at 7:00 AM (Central time) and drove 52 miles to the Dallas airport at an average speed of 56 mph. Two hours later, her plane flew 1095 miles from Dallas to Idaho Falls at an average speed of 355 mph. Thirty minutes later, she drove a rental car 64 miles from Idaho Falls to Aberdeen at an average speed of 68 mph. What time did she arrive in Aberdeen? (Mountain time)**

   (A) 1:21 PM (B) 1:24 PM (C) 1:27 PM (D) 1:30 PM (E) 1:33 PM

4. **Cindy can paint a classroom in 5 hours while Scott needs 6 hours to paint a classroom. If they work together, how long will it take them to paint 12 classrooms? (nearest minute)**

   (A) 32 hr 41 min (B) 32 hr 44 min (C) 32 hr 47 min (D) 32 hr 50 min (E) 32 hr 53 min

5. **The formula for the moment of inertia of a hollow sphere is $I = \frac{2}{3}mr^2$, where I is the moment of inertia in kilograms × (meters)^2, m = mass in kilograms, and r = radius in meters. If a hollow sphere has a moment of inertia of 42.592 kilograms × (meters)^2 and a mass of 3.3 kg, what is the diameter of the sphere? (nearest tenth)**

   (A) 8.4 m (B) 8.6 m (C) 8.8 m (D) 9.0 m (E) 9.2 m

6. **Farmer Fred has a pond with catfish, bass and crappie. There are 288 fish in the pond. The number of catfish is 18 more than the number of crappie and the number of crappie is 12 less than three times the number of bass. How many bass are in the pond?**

   (A) 36 (B) 38 (C) 40 (D) 42 (E) 44

7. **Consider the function $f(x) = \frac{3-4x}{5-6x}$. If $h(x)$ is the inverse function of $f(x)$ and $h(c) = 0.875$, then c = ________. (nearest tenth)**

   (A) 1.4 (B) 1.6 (C) 1.8 (D) 2.0 (E) 2.2

8. **Cooper lives 3 miles from Argyle High School. He can run to school in 20 min 30 sec or he can walk to school in 48 min 30 sec. One day, he left home and ran for 2.25 miles and walked the rest of the way. How long did it take him to get to school? (nearest second)**

   (A) 27 min 2 sec (B) 27 min 9 sec (C) 27 min 16 sec (D) 27 min 23 sec (E) 27 min 30 sec

9. Find the perimeter of pentagon ABCDE.
(nearest tenth)

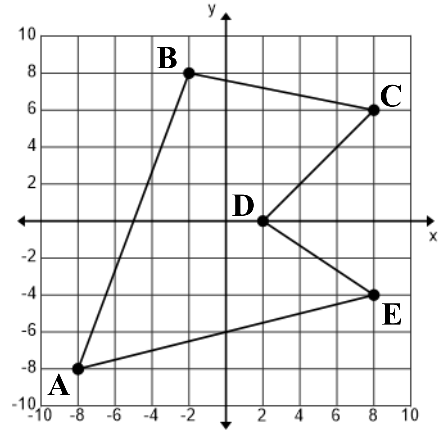
(A) 58.3 (B) 58.6 (C) 58.9

(D) 59.2 (E) 59.5

10. Find the area of pentagon ABCDE.
(nearest whole number)

(A) 128 (B) 130 (C) 132

(D) 134 (E) 136



Problems 9, 10, 11, 12

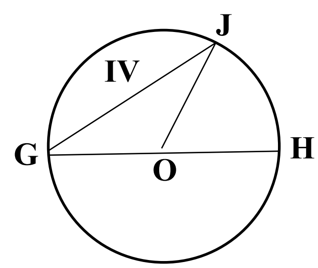
11. m∠DEA = ________°. (nearest tenth)

(A) 47.7 (B) 48.0 (C) 48.3 (D) 48.6 (E) 48.9

12. Consider points $G(12,-16)$ and $H(20,b)$. If $\overleftrightarrow{GH}$ is parallel to $\overleftrightarrow{AE}$, then b = _______.

(A) −14 (B) −13 (C) −12 (D) −11 (E) −10

13-15. Consider the circle on the right with center O and diameter $\overline{GH}$. The measure of minor arc $\overset{\frown}{GJ} = 116°$ and GH = 26.



Problems 13, 14, 15

13. The area of sector JOH is _______. (nearest whole number)

(A) 90 (B) 92 (C) 94 (D) 96 (E) 98

14. The perimeter of triangle GOJ is _______. (nearest whole number)

(A) 47 (B) 48 (C) 49 (D) 50 (E) 51

15. Region IV is enclosed by minor arc $\overset{\frown}{GJ}$ and chord $\overline{GJ}$. Find the area of region IV.
(nearest whole number)

(A) 91 (B) 92 (C) 93 (D) 94 (E) 95

16. A right circular cone has a radius of 5.75 and a total surface area of 331. Find the volume of the cone.
(nearest whole number)

(A) 378 (B) 381 (C) 384 (D) 387 (E) 390

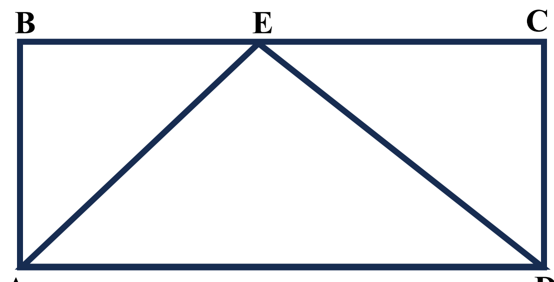
17. The area of equilateral triangle ABC equals the area of circle O. If the diameter of circle O = 16.44, then the perimeter of triangle ABC = __________. (nearest whole number)

(A) 62 (B) 64 (C) 66 (D) 68 (E) 70

18. Consider ΔABC with $m\angle B = 90°$. Point D lies on $\overline{AC}$ with $\overline{AC} \perp \overline{BD}$, AD = 7.4 and BD = 9.2. The perimeter of ΔABC = ________. (nearest tenth)

(A) 44.7 (B) 45.0 (C) 45.3 (D) 45.6 (E) 45.9

19-20. Consider rectangle ABCD and isosceles triangle AED. AB = 24 and the area of rectangle ABCD is 1056.



Problems 19, 20

19. Find the perimeter of ΔAED. (nearest whole number)

(A) 107 (B) 109 (C) 111

(D) 113 (E) 115

20. Consider point P that lies on $\overline{AE}$ such that $\overrightarrow{DP}$ bisects $\angle ADE$. Find the length of $\overline{PE}$. (nearest tenth)

(A) 13.8 (B) 14.1 (C) 14.4 (D) 14.7 (E) 15.0

21. Consider an arithmetic sequence in which the seventh term is 31 and the sixteenth term is 67. What is the product of the twenty-fifth and twenty-sixth terms?

(A) 11,017 (B) 11,019 (C) 11,021 (D) 11,023 (E) 11,025

22. If the area of the circle $x^2 + y^2 + 10x - 14y + f = 0$ is 121π, then f = ________.

(A) -47 (B) -45 (C) -43 (D) -41 (E) -39

23. Three of the zeros of the fourth-degree polynomial $x^4 + bx^3 + cx^2 + dx + e$ are -4, 3, and $1+\sqrt{7}$. $c + d =$ ________.

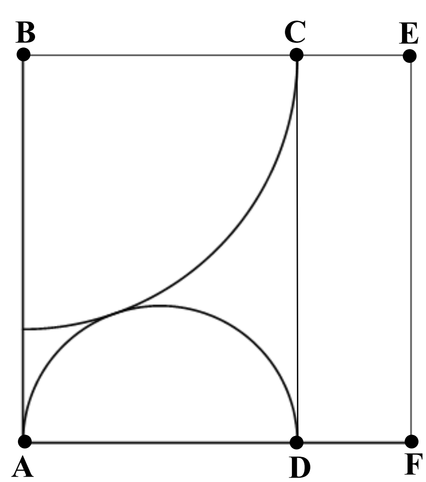
(A) -4 (B) -2 (C) 0 (D) 2 (E) 4

24. Consider the sequence $72, 60, 50, 41.\overline{6}, 34.7\overline{2}, \ldots$ The sum of the first 10 terms is ________. (nearest hundredth)

(A) 361.79 (B) 361.90 (C) 362.01
(D) 362.12 (E) 362.23

25. Given: Rectangle ABCD, Square ABEF, EF = 4. Quarter circle is tangent to semicircle. Area of quarter circle plus area of semicircle = ________. (nearest hundredth)

(A) 9.36
(B) 9.39
(C) 9.42
(D) 9.45
(E) 9.48



Problem 25

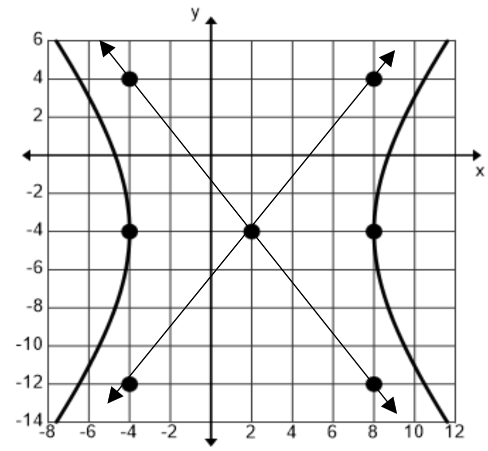
26. Angle A is in quadrant IV and angle B is in quadrant II. If $\cos A = \frac{12}{13}$ and $\sin B = \frac{24}{25}$, then $\tan(A+B) =$ _________.

(A) $\frac{319}{36}$ (B) $\frac{80}{9}$ (C) $\frac{107}{12}$ (D) $\frac{161}{18}$ (E) $\frac{323}{36}$

27. Find the obtuse angle formed by the two intersecting lines shown on the right. (nearest whole number)

(A) 102° (B) 103° (C) 104°

(D) 105° (E) 106°



Problems 27, 28

28. The two lines are the asymptotes of a hyperbola. The equation of the hyperbola is of the form $\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$. Given: a and b are integers with $a < b < 9$. One of the foci is the point $(h+c, k)$. $h+c+k =$ ________. (nearest whole number)

(A) 4 (B) 6 (C) 8 (D) 10 (E) 12

29. Charisa is standing on a platform at the water's edge as she watches the USS Oscar Austin, a Navy destroyer, sail directly away from where she is watching on a platform. The high point of the destroyer is 208 feet above the waterline and Charisa's eye level is 60 feet above the waterline. How far is the high point of the destroyer from Charisa just as the high point of the destroyer disappears from her sight? The radius of the earth is 3960 miles. (nearest hundredth)

(A) 26.82 mi (B) 26.93 mi (C) 27.04 mi (D) 27.15 mi (E) 27.26 mi

30. Consider the sequence $3, 4, 7, 10, 16, 21, 30, 40, 57, \ldots$ The sum of the next two terms is __________.

(A) 201 (B) 202 (C) 203 (D) 204 (E) 205

31. Planet Z has 72-hour days. In the month of Tam, the temperature varies sinusoidally with a low of 28°F at t = 0 hr and a high of 88°F at t = 36 hr. On a typical Tam day, the temperature is equal to or above 42°F for _________ hours. (nearest tenth)

(A) 48.6 (B) 48.9 (C) 49.2 (D) 49.5 (E) 49.8

32. The vertices of a triangle are $(-6, 10)$, $(8, k)$, and $(k, -12)$. The area of the triangle is 124. If k > 0, then k = ____.

(A) 2 (B) 4 (C) 6 (D) 8 (E) 10

33. A Navy jet leaves an aircraft carrier and flies due west at 660 mph. The carrier continues to travel 55° south of west at 22 mph. If the jet has enough fuel to fly for 6 hours, how far west can the jet travel before it must turn and return to the carrier? (Ignore the curvature of the earth) (nearest mile)

(A) 2016 mi (B) 2019 mi (C) 2022 mi (D) 2025 mi (E) 2028 mi

34. Consider a set of consecutive positive odd integers beginning with 1 and ending with n. If one of the integers is removed, the average of the remaining integers is $33\frac{1}{4}$. Which integer is removed?

(A) 17 (B) 19 (C) 21 (D) 23 (E) 25

35. Des has two samples of radioactive materials. The mass of sample A is 2.4 g and it has a half-life of 120 days. The mass of sample B is 3.6 g and it has a half-life of 96 days. How long will it take for the samples to decay to equal amounts? (nearest hundredth)

(A) 278.76 days (B) 279.77 days (C) 280.78 days (D) 281.79 days (E) 282.80 days

36-37. An NFL quarterback released a pass at a height of 7 feet above the ground. A receiver caught the football at a height of 5 feet above the ground, 45 yards downfield from the quarterback. The pass was released at an angle of 20° above the horizontal. The acceleration of gravity = 32.174 ft/s^2.

36. How much time elapsed from when the football was released and when it was caught? (nearest hundredth)

(A) 1.75 sec (B) 1.78 sec (C) 1.81 sec (D) 1.84 sec (E) 1.87 sec

37. The speed of the football as it was released was __________ mph. (nearest whole number)

(A) 53 (B) 54 (C) 55 (D) 56 (E) 57

38. Given two points with Cartesian coordinates $T(9, t)$ and $R(9, r)$, $t > 0$ and $r > 0$. Consider the graph of the hyperbola produced by the polar equation $r = \dfrac{16}{3 + 5\cos\theta}$ and containing point T. Point R lies on one of the asymptotes of the hyperbola. $r - t =$ ____________________. (nearest tenth)

(A) 1.4 (B) 1.5 (C) 1.6 (D) 1.7 (E) 1.8

39. The graph of $x^2 + 2xy + y^2 - 1 = 0$ produces two parallel lines. Find the distance between the lines. (nearest hundredth)

(A) 1.41 (B) 1.56 (C) 1.71 (D) 1.86 (E) 2.01

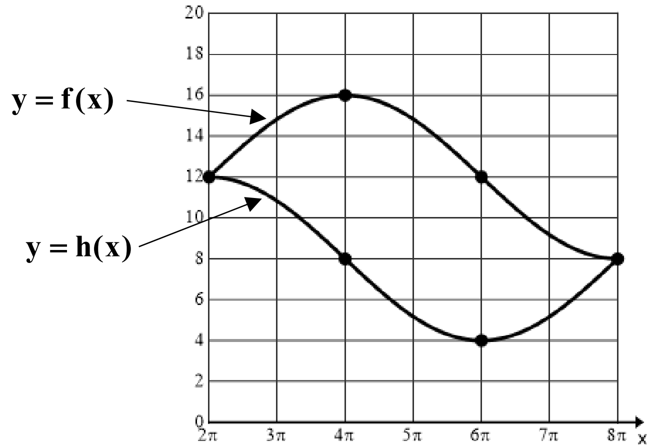
40. Sri rolls two six-sided dice and calculates the product of the top faces. What is the probability that the product is divisible by 4? (nearest ten-thousandth)

(A) 0.4161 (B) 0.4164 (C) 0.4167 (D) 0.4170 (E) 0.4173

41-44. Given: $f(x)=12-4\cos\left(\dfrac{x}{4}\right)$ and $h(x)=8+4\sin\left(\dfrac{x}{4}\right)$. Let R be the region enclosed by the graphs of $y=f(x)$ and $y=h(x)$ over the closed interval $[2\pi, 8\pi]$.

41. Determine the number c that satisfies the conclusions of the mean value theorem for the function f(x) on $[4\pi, 6\pi]$. (nearest hundredth)

(A) 15.30 (B) 15.33
(C) 15.36 (D) 15.39
(E) 15.42



Problems 41, 42, 43, 44

42. Find the area of the region R. (nearest whole number)

(A) 95 (B) 98
(C) 101 (D) 104
(E) 107

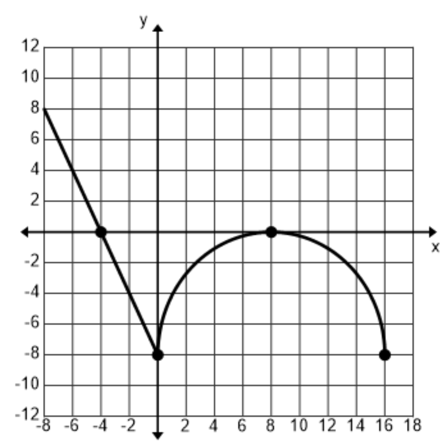
43. Find the volume of the solid generated by revolving region R about the line $y=-2$. (nearest whole number)

(A) 8086 (B) 8089 (C) 8092 (D) 8095 (E) 8098

44. The arc length of the graph of the curve $y=h(x)$ from $x=3\pi$ to $x=b$ equals 10.263661. $b=$_________. (nearest hundredth)

(A) 17.24 (B) 17.28 (C) 17.32 (D) 17.36 (E) 17.40

45-46. The domain of $f(x)$ is $-8\le x\le 16$. The graph of $f'(x)$ consists of the line segment and the semicircle shown. $f(-6)=10$.



Problems 44, 45, 46

45. The absolute maximum of $f(x)$ is ________. (nearest whole number)

(A) 12 (B) 13 (C) 14
(D) 15 (E) 16

46. The absolute minimum of $f(x)$ is _________. (nearest hundredth)

(A) -29.47 (B) -29.36 (C) -29.25
(D) -29.14 (E) -29.03

47. Consider the graph of $x^2y^2 - 4x^2 - 3y^2 = 0$. Find the equation of the line tangent to this graph at the point $P(3, b)$, $b > 0$. The y-intercept of this tangent line is the point $Q(0, d)$. $d =$ ________. (nearest hundredth)

(A) 3.58 (B) 3.61 (C) 3.64 (D) 3.67 (E) 3.70

48. Consider the curve given by the parametric equations $x(t) = 3t^3 - 2t^2$ and $y(t) = t^3 - 6t$. Find the value of $\frac{d^2y}{dx^2}$ when $t = 1$. (nearest thousandth)

(A) 0.570 (B) 0.573 (C) 0.576 (D) 0.579 (E) 0.582

49-50. A particle moves along the x-axis in such a way that its position at any time t is given by $x(t) = 2t^4 - 15t^3 + 20t^2 + t,\ 0 \le t \le 5$.

49. Find the maximum speed of the particle during this time interval. (nearest whole number)

(A) 64 (B) 67 (C) 70 (D) 73 (E) 76

50. Find the total distance traveled by the particle from $t = 0$ to $t = 5$. (nearest whole number)

(A) 162 (B) 165 (C) 168 (D) 171 (E) 174

51. The rate at which a baby Great Dane gains weight is proportional to the difference in its adult weight and its current weight given by $\frac{dw}{dt} = \frac{1}{20}(100 - w)$. Let $w(t)$ be the weight in pounds at time t in weeks. If a Great Dane puppy weighed 5 pounds at birth, find the weight of the Great Dane at 64 weeks. (nearest whole number of pounds)

(A) 92 (B) 93 (C) 94 (D) 95 (E) 96

52. Find the area of the region enclosed by the inner loop of the graph of the polar equation $r = 1 - 2\cos\theta$ for $0 \le \theta \le 2\pi$. (nearest hundredth)

(A) 0.54 (B) 0.57 (C) 0.60 (D) 0.63 (E) 0.66

53-54. Assume that the distribution of the weights of Coastal Brown Bears is approximately normal with a mean of 1,000 pounds and a standard deviation of 50 pounds.

53. What proportion of Coastal Brown Bears weigh more than 965 pounds? (nearest thousandth)

(A) 0.725 (B) 0.736 (C) 0.747 (D) 0.758 (E) 0.769

54. If six Coastal Brown Bears are randomly selected, what is the probability that at least four of the bears will weigh more than 965 pounds? (nearest thousandth)

(A) 0.799 (B) 0.810 (C) 0.821 (D) 0.832 (E) 0.843

Scores	322	334	315	322	336

55. Kevin Burnett participated in the Calculator Applications Tournament of Champions. Each contestant took five extremely difficult tests written by Don Kirby. Kevin's results are in the table above. Find the sum of the mode, mean, median and range of Kevin's scores. (nearest tenth)

(A) 987.2 (B) 988.4 (C) 989.6 (D) 990.8 (E) 992.0

56. David and Trevor both run 30 minutes every morning before school. The distance David runs is approximately normally distributed with a mean of 4.2 miles and a standard deviation of 0.44 miles. The distance Trevor runs is approximately normally distributed with a mean of 4.0 miles and a standard deviation of 0.33 miles. The distance David runs each morning is independent of the distance Trevor runs. Randomly select one morning before school. Let the random variable X be the difference in distance each boy runs. $X = D - T$. Find the probability that Trevor runs farther than David. (nearest thousandth)

(A) 0.342 (B) 0.346 (C) 0.350 (D) 0.354 (E) 0.358

57. Bottles of Cream Soda are supposed to contain 20.0 ounces of soda. From past quality control data, the distribution of the contents is approximately normal. An inspector suspects the bottles are being underfilled recently. He measures the contents of six randomly selected bottles that were filled on Monday. The results in ounces are 19.8, 19.7, 20.0, 19.9, 19.7 and 20.0. Does this data provide convincing evidence that the bottles are filled with less than 20.0 ounces? To find out, the inspector performed an appropriate test and obtained a p-value of _________. (nearest thousandth)

(A) 0.022 (B) 0.033 (C) 0.044 (D) 0.055 (E) 0.066

Planet	Distance from the Sun (AU)	Period (earth years)
Venus	0.723	0.615
Mars	1.524	1.881
Jupiter	5.203	11.862
Saturn	9.539	29.456
Neptune	30.061	164.810

58-59. Dr. Stat plotted the data and noticed a curved pattern.

58. Which of the following plots would be approximately linear? (Period = P, Distance = D)

(A) P vs ln(D) (B) ln(P) vs D (C) ln(P) vs ln(D) (D) P vs D^2 (E) P vs $D^{2.5}$

59. Use a transformation to linearize the data. Find the LSRL of the transformed data. If the period of Uranus is 84.07 earth years, what is the distance from the Sun to Uranus? (nearest hundredth)

(A) 19.19 AU (B) 19.22 AU (C) 19.25 AU (D) 19.28 AU (E) 19.31 AU

60. Mr. Speir believes that the class 5A region II mathematics contest will be tough this year. He predicts that the distribution of scores will be approximately normal with a mean of 220 and a standard deviation of 50.7. He also predicts that the winner will score at the 98^{th} percentile and will answer 58 questions. If Mr. Speir is correct, how many questions will the region champion miss?

(A) 0 (B) 1 (C) 2 (D) 3 (E) 4

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University Interscholastic League
MATHEMATICS CONTEST
HS ● District ● 2026
Answer Key

1. E	21. C	41. B
2. B	22. A	42. E
3. C	23. B	43. E
4. B	24. E	44. B
5. C	25. C	45. C
6. D	26. E	46. A
7. D	27. E	47. D
8. E	28. C	48. C
9. E	29. D	49. E
10. E	30. D	50. D
11. A	31. B	51. E
12. A	32. B	52. A
13. C	33. A	53. D
14. B	34. E	54. E
15. E	35. C	55. D
16. D	36. B	56. E
17. C	37. C	57. A
18. C	38. E	58. C
19. B	39. A	59. A
20. A	40. C	60. D